Graph Theory

Lecture 4

Connectivity, Disjoint Paths & Network Robustness

From Lecture 3 to Lecture 4

Last lecture: Which single vertex or edge can disconnect a network?

- Cut vertex → vulnerable vertex
- Bridge → vulnerable edge

Today: How many vertices or edges must fail before the network disconnects?

This moves us from identifying vulnerability to measuring robustness.

Learning Outcomes

Determine vertex and edge cuts.

Calculate vertex connectivity $\kappa(G)$ and edge connectivity $\lambda(G)$.

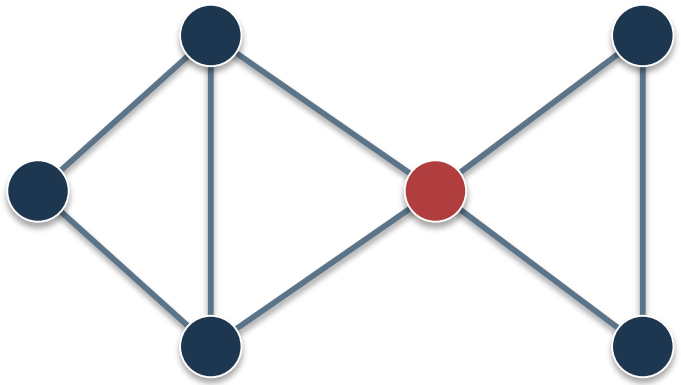
Relate $\kappa(G)$, $\lambda(G)$, and minimum degree $\delta(G)$.

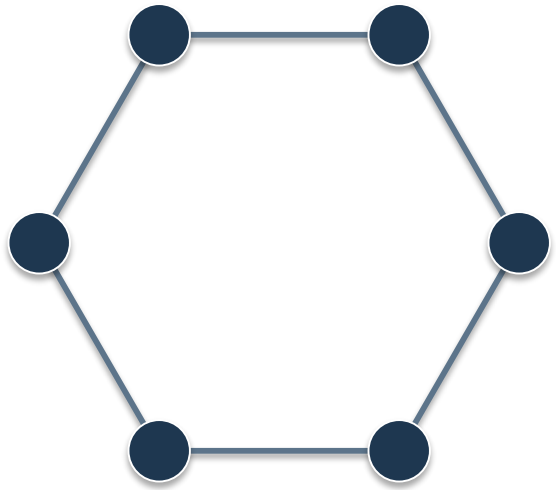
Distinguish vertex-disjoint and edge-disjoint paths.

Explain and apply the central idea of Menger's theorem.

Interpret connectivity measures as fault tolerance in real networks.

Real-Life Question: Which Network Is More Reliable?





A: one critical router can separate it

B: one router failure does not disconnect it

Vertex Cut / Separating Set

A vertex cut is a set of vertices whose removal disconnects the graph.

A cut vertex is a vertex cut containing only one vertex.

Real-world interpretation: routers, airports, substations or servers that must fail together to split a network.

We seek the smallest such set.

Vertex Connectivity κ(G)

κ(G) = minimum number of vertices whose removal disconnects G.

Interpretation: resistance to vertex failures.

- κ(G)=1 → a cut vertex exists.
- Larger κ(G) → greater vertex redundancy.

For complete graphs, $\kappa(K_n)=n-1$ under the standard convention.

Edge Cut

An edge cut is a set of edges whose removal disconnects the graph.

A bridge is an edge cut of size 1.

Real-life interpretation: fiber links, roads, power lines or communication channels.

We seek the smallest number of link failures that disconnects the system.

Edge Connectivity λ(G)

λ(G) = minimum number of edges whose removal disconnects G.

Interpretation: resistance to link failures.

- λ(G)=1 → the graph contains a bridge.
- Larger λ(G) → more edge redundancy.

Examples: $\lambda(P_n)=1$, $\lambda(C_n)=2$, $\lambda(K_n)=n-1$.

Vertex vs Edge Connectivity

κ(G): minimum VERTEX failures.

λ(G): minimum EDGE failures.

Same network, two different reliability questions:

- How many routers/devices must fail?
- How many communication links must fail?

A Fundamental Relationship

$\kappa(G) \leq \lambda(G) \leq \delta(G)$

$\delta(G)$ is the minimum degree of G.

Why $\lambda(G) \leq \delta(G)$? Remove all edges incident to a minimum-degree vertex; that vertex becomes isolated.

The inequality connects global robustness with local degree.

Independent Routes: Disjoint Paths

Two u–v paths are edge-disjoint if they share no edges.

Two u–v paths are internally vertex-disjoint if they share no internal vertices.

They may share endpoints u and v.

Real-world meaning: alternative routes that do not depend on the same links or intermediate facilities.

Menger's Theorem: The Main Idea

Independent routes $\leftrightarrow$ minimum separating failures.

Vertex version: maximum internally vertex-disjoint u–v paths equals the minimum size of a u–v vertex separator (for distinct nonadjacent u,v).

Edge version: maximum edge-disjoint u–v paths equals the minimum number of edges separating u and v.

Think of Menger's theorem as a mathematical statement of redundancy.

Menger's Theorem: Keep the Idea Simple

Number of independent routes = minimum failures needed to separate

If there are 3 independent routes between two places, one failure cannot block communication.

To disconnect them, every independent route must be broken.

Menger's theorem formalizes this redundancy idea.

We will use the concept more than the formal proof in this lecture.

Real-Life Interpretation: Alternative Routes

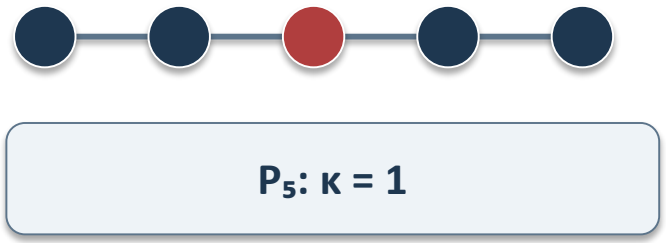
Suppose two cities or data centers are connected by three independent routes.

If no two routes share an intermediate junction, there are three internally vertex-disjoint paths.

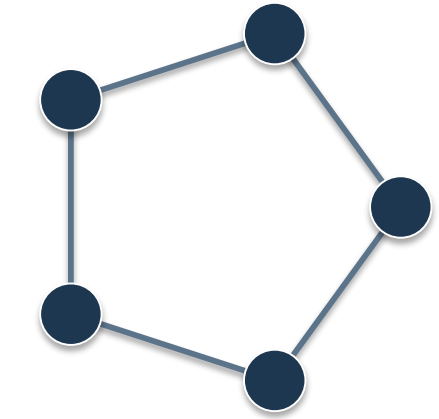
More independent routes mean stronger fault tolerance.

Graph theory turns 'redundancy' into something measurable.

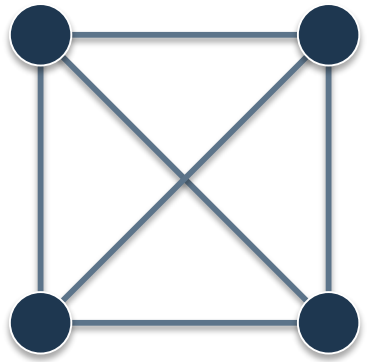
Examples: κ(G) for Familiar Graphs



P_5: κ = 1



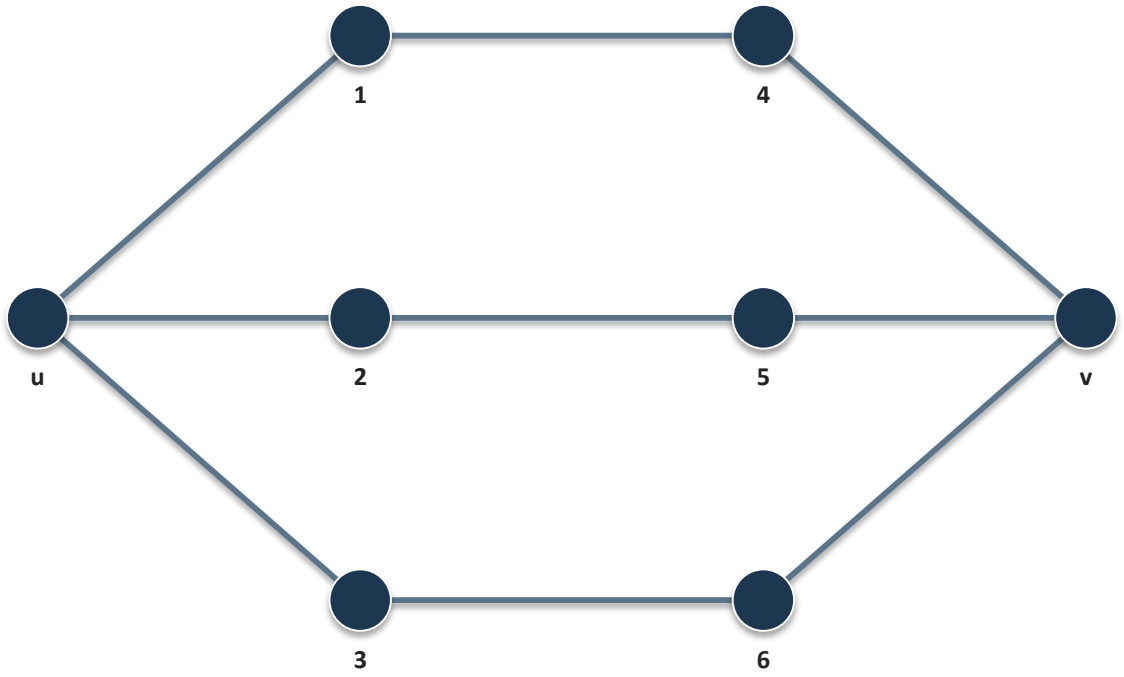
C_5: κ = 2



K_4: κ = 3

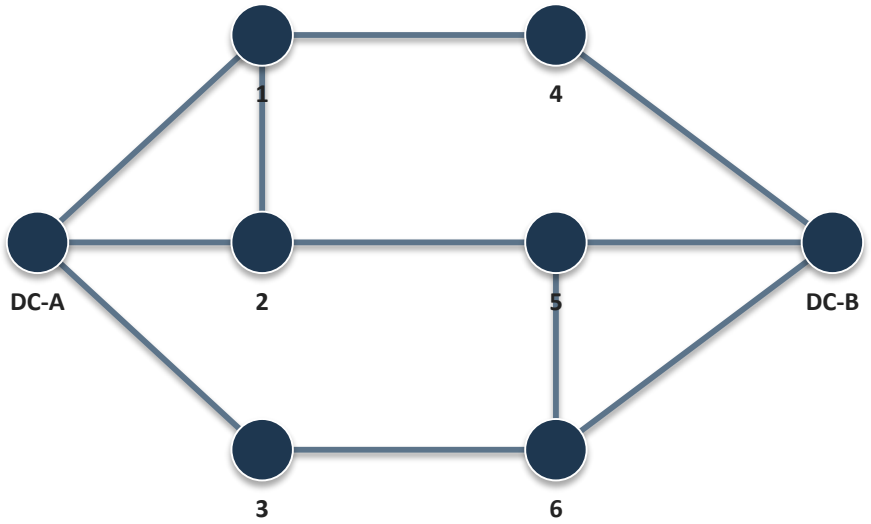
Higher κ means more vertex failures are required to disconnect the graph.

Pictorial Example: Three Independent Routes



Three internally vertex-disjoint u–v paths

Applied Problem: Data-Center Connectivity



Determine:

- How many independent routes visibly connect DC-A and DC-B?
- Find an edge cut.
- Find a vertex separator.
- What does each answer mean operationally?

Concept Check

1. If a graph has a cut vertex, what is $\kappa(G)$?
2. If a graph has a bridge, what is $\lambda(G)$?
3. What does $\kappa(G) \leq \lambda(G) \leq \delta(G)$ tell us?
4. Edge-disjoint vs internally vertex-disjoint paths?
5. Why does a cycle improve reliability?
6. Which is more robust against vertex failures: P_6 or C_6?

Mini Design Challenge

A company requires communication between two offices to survive any single link failure.

What graph-theoretic property is required?

How many edge-disjoint routes are needed at minimum?

Extension: what changes if the system must survive any single intermediate-router failure?

Translate the engineering requirement into a graph requirement before drawing the network.

Mini Design Challenge

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Class Quiz + Design Challenge

Attempt individually or in pairs, as instructed.

Keep answers brief.

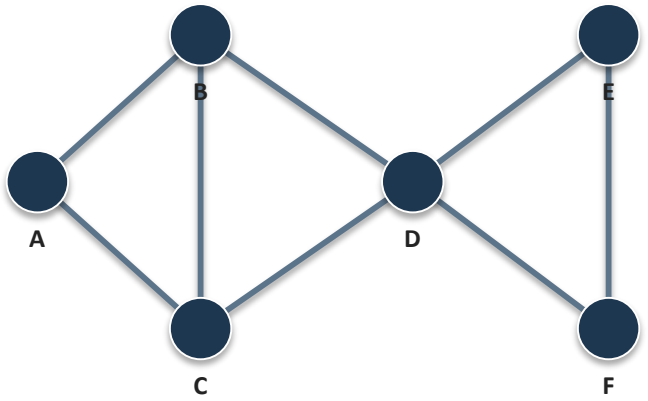
Use graph reasoning; long formal proofs are not required.

Marks distribution:

- Quick review questions: 4 marks
- Connectivity question: 2 marks
- Design challenge: 4 marks

Quiz Part A — Quick Review Questions

4 marks, about 6 minutes



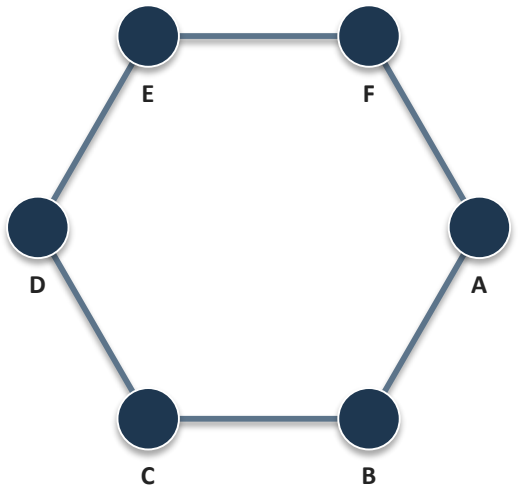
Q1. In the given graph, identify:

- one cut vertex
- one bridge, if any

Q2. Two graphs have the same degree sequence. Is this enough to conclude that they are isomorphic? Give one-line justification.

Marks: Q1 = 2, Q2 = 2

Quiz Part B — Connectivity Question



Q3. For the cycle network shown:

What is the minimum number of vertices that must be removed to disconnect it?

What is the minimum number of edges that must be removed to disconnect it?

Write the values of $\kappa(G)$ and $\lambda(G)$.

Quiz Part C — Design Challenge



Current network has only one route from Office A to Office B.

Requirement: A and B must remain connected if any one communication link fails.

Task:

- Add the minimum new connection(s).
- Draw the modified network.
- Explain in one sentence why it now survives one link failure.

Expected idea: create an alternative route / cycle.

Lecture Summary

Vertex cuts generalize cut vertices; edge cuts generalize bridges.

$\kappa(G)$ measures resistance to vertex failures; $\lambda(G)$ measures resistance to edge failures.

$\kappa(G) \leq \lambda(G) \leq \delta(G)$ relates global robustness to local degree.

Disjoint paths represent independent alternatives.

Menger's theorem links independent routes with minimum separating failures.

Next: Advanced Trees and Spanning Trees.